GLOBAL INSTITUTE OF TECHNOLOGY
B.TECH I- SEM MAIN EXAM ,JAN 2025 (GUESS PAPER)
SUBJECT -1FY1-02 ENGINEERING PHYSICS-2025

PART-A (2 Marks Each question)

1. What do you mean by resolving power of an optical instrument?
2. What is normalized and orthogonal wave function?
3. Explain total internal reflection?
4. What is the relation between Einstien's Cofficients? Explain them.
5. What is hall effect?
6. What is scalar and vector field?
7. Define curl and divergence of a vector.
8. What do you mean by Spectral purity?
9. What is significance of wave function?
10. Explain how X-ray diffraction can be used to characterize nano particles?
11. What is meant by numerical aperture for an optical fibre?
12. Sketch a neat diagram of Fraunhoffer diffraction at a single slit.
13. State the Maxwell's equations in their differential form.
14. Define matter waves.
15. Why newton's rings are circular in shape?
16. Write Expression for Fermi-Dirac distribution function?
17. Mention two properties of metallic bond?
18. Write two differences between spatial and temporal coherence?
19. State physical significance of curl of static magnetic field.
20. Express brag's condition of X-ray diffraction?
21. Write all four Maxwell's equations in integral form for free space.
22. State Rayleigh's criterion of resolution.
23. What do you mean by stimulated emission and spontaneous emission?
24. The carrier concentration in n-type semiconductor 10^{19} per m^3. what is the value of Hall cofficient?
25. What do you mean by eigen functions and eigen values?
26. Explain the advantages of optical fibres in communucation?
27. When the movable mirror of Michelson's interferometer is shifted by 0.030 mm, the shift of 100 fringes is observed. Calculate the wavelength of light in A^o and stste its colour.
28. What will be the effect on diameters in Newton's ring experiment if film is of μ refractive index?
29. Write expressions of Bio-savart's law in vector form.
30. State the essential difference between interference and diffraction of light.

PART-B (4 Marks each question)

Q1. Explain the essential requirement for production of laser action.

Q.2 Derive an expression for hall Coefficient. Mention two applications of Hall effect.

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Q.3 deduce the expression for poynting vector and explain its physical meaning.

Q.4 Michelson interferometer experiment is performed with a source which have two wavelength 5882 A° and 5886 A°. By what distance, does the mirror have to be move between two positions of disappearance of fringes?

Q.5 Derive time independent Schrodinger's wave equation.

Q.6 Two wave-trains overlaps 40% of their length. If the maxima in the resulting interference pattern receives 20 units of light, how much do the minima receives?

Q.7 A parallel beam of sodium light is incident normally on a plane transmission grating having 4250 lines /cm and a second order spectral line is observed at an angle 30°. Determine the wavelength of light.

Q.8 Give the construction and theory of plane transmission grating and explain the formation of spectra by it.

Q.9A ray of light entres from air into fiber. The refractive index of air is one. The fiber has a core of refractive index 1.5 and cladding of refractive index 1.48. Find the critical angle, the fractional refractive index, acceptance angle and numerical aperture.

Q.10 A plane transmission grating of length 6 cm has 5000 lines/cm. Find the resolving power of grating and the smallest wavelength difference that can be resolved for light of wavelength 5000 Å.

Q.11 Calculate the angles at which the first dark band and the next bright band are formed in the Fraunhofer diffraction pattern of a slit 0.3 mm wide (λ =5890 A°) .

Q.12A single slit is illuminated composed of two wavelengthsλ1 and λ2 One observes that due to diffraction, the first minima obtained for λ1 coincides with the second diffraction minima of lambda λ2 What will be the relation between λ1 and λ2?

Q.13 A laser beam has a power of 50 mw. It was an aperture of $5X10^{-3}$ m and wavelength 7000 Å. A beam is focused with a tens of focal length 0.2 m. Calculate the area spread and intensity of the image.

Q.14 An optical fibre has a numerical aperture of 0.2 and cladding refractive index of 1.59. Determine the acceptance angle for the fiber in water which has a refractive index of 1.33.

Q.15 Find the probability that a particle is in one dimensional box of length I can be found between 0.45 1 and 0.55 1 for the ground and first excited states.

Q.16 Explain how you can distinguish between linearly, circularly and elliptically polarized light.

Q.17 Explain how Newton's rings experiment can be used to find the wavelength of the light source used.

Q.18 What is de Broglie's hypothesis? Calculate the de-Broglie wavelength associated with electrons, which are accelerated by a voltage of 50 kV.

Q.19 Explain top down and bottom up approach for the synthesis of nano materials.

Q.20 Calculate the Zero-point energy for a particle in an infinite potential well for an electron confined to a 1 nm atom.

Q.21 Explain the principle and working of a He-Ne laser by clearly drawing the energy level diagrams.

Q.22 What is Hall effect and its importance in semiconductor industry?

PART -C (10 marks of Each question)

Q.1 What are orthogonal wave functions? Write the Schrodinger equation for particle in a box and solve it to obtain energy Eigen values and Eigen functions.

Q.2(a) Explain what is population inversion and pumping in lasers?

(b) With neat diagrams, describe the principle, construction and working of semiconductor laser. Also,
Q.3 (a) What is diffraction grating?Describe how the wavelength of the monochromatic light is determined using it.

(b) Find the number of orders visible if the wavelength of the incident radiation is 5000 A and number of lines on the grating are 1000 in one centimeter.determined using it.

Q.4(a) How would you produce polarized light by reflection? What is Brewster's Law? Calculate the angular position of the sun above the horizon so that light reflected from a galm lake is completely polarized. The refractive index of water is 1.33.)

(b) How would you ascertain the correct state of polarization (i.e. circularly, elliptically, plane, unpolarized, partial polarized etc.) of an unknown incoming beam of light with the help of a rotating Nicol prism and a quarter wave plate?

Q.5 (a) Why semiconductors are doped? Giving suitable energy level diagram explain how doping by donors improve conductivity of a semiconductor.

(b) Explain Meissner's effect.

Q.6 Describe and explain the formation of Newton's rings in reflected monochromatic light. How can these be used to determine the wavelength of light? Derive the formula used.

Q.7 (a) Derive the Schrodinger time dependent equation and explain the physical meaning of wave function Ψ.

(b) What do you mean by degeneracy?

Q.8 (a) Discuss the formation of energy bands in solids.

(b) Classify the solids on the basis of energy bands and discuss the conductivity in semiconductors.

Q.9 Derive the formula for curl and divergence for electrostatic field and static magnetic field.

Q.10 (a) What is an optical fibre? Obtain an expression for numerical aperture of step index optical fibre.

(b) Explain visibility of fringes as a measure of coherence.

Q.11Describe Fraunhofer diffraction due to a single slit. Deduce the position of maximas and minimas.

Q.12 Show that energy of an electron confined in a ID potential well of length 'L' and infinite depth is quantized. Is the electron allowed to have zero energy? Comment.

Q.13 What are the basic requirements of semi-conductor laser? With the help of energy band diagram explain working of semi-conductor laser.

Q.14 Deduce the expressions for Maxwell's Equations in integral and differential form. Also discuss their physical significance.

Q.15 Explain the terms: Population inversion and optical pumping. Discuss with suitable diagrams the principle, construction and working of Helium-Neon Laser.

Q.16 State and prove Poynting theorem for the rate of flow of energy in electromagnetic field. What is Poynting vector?

Q.17 Give physical significance of wave function. Derive time dependent and time independent Schrödinger wave equation.